

ORIGINAL RESEARCH OPEN ACCESS

Academic Integrity Within the Medical Curriculum in the Age of Generative Artificial Intelligence

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Received: 25 August 2024 | **Revised:** 24 January 2025 | **Accepted:** 7 February 2025

Funding: The authors received no specific funding for this work.

Keywords: academic integrity | artificial intelligence | medical education | plagiarism detection | project-based learning

ABSTRACT

Background: The integration of generative artificial intelligence (AI) technologies, such as GPT-3, Wordtune, and Jenni, into academic settings has revolutionized content creation, raising significant questions about authorship and originality. While AI offers benefits in efficiency and productivity, it presents substantial challenges to academic integrity. This paper examines these challenges and the need for new frameworks and policies to ensure ethical AI use.

Materials and Methods: Our study aims to develop frameworks promoting ethical AI use and safeguarding academic work authenticity. We employed project-based learning (PBL) methodology to enhance student engagement and academic performance. A total of 179 students participated. PBL involves students in real-world projects, fostering critical thinking and research skills. Turnitin was used to evaluate the similarity percentage of submitted research papers, with a maximum allowable similarity of 20%.

Results: Results showed that PBL students had higher academic performance (82.5 vs. 66.5) and lower similarity percentages (4.5% vs. 13%) compared to traditional literature review assessments. The use of Turnitin effectively identified AI-generated content, although it struggled with more sophisticated texts.

Conclusion: Our findings highlight the effectiveness of PBL in promoting originality and reducing AI-generated plagiarism. Integrating these methodologies, along with ethical AI education, can help institutions maintain academic integrity while leveraging AI benefits. Future research should refine these strategies to adapt to the evolving educational landscape.

1 | Introduction

The integration of generative artificial intelligence (AI) technologies, such as GPT-3, Wordtune, Jenni, and so forth, into academic settings [1] has revolutionized the way students and researchers produce written content. These tools can generate high-quality text with minimal human input [2, 3], raising significant questions about the nature of authorship and originality. While AI offers considerable benefits in terms of efficiency and productivity [4], it also presents substantial challenges to academic integrity [5]. This paper examines these

challenges and the necessity for new frameworks and policies to ensure ethical AI use in academic settings.

Generative AI tools have the capability to produce coherent [6], contextually relevant text that is often indistinguishable from human-authored content [7]. This phenomenon, known as augmented writing, enables AI to assist in generating entire essays or substantial parts of academic papers [8]. This capability blurs the line between human and machine authorship, raising fundamental questions about the true origin of ideas and the role of the author in the creation process [9]. Furthermore,

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the widespread use of these AI tools has led to an increase in AI-assisted submissions in academic settings, complicating the detection of plagiarism [10]. Traditional plagiarism detection tools often struggle to differentiate between human and AI-produced content [11]. This difficulty is exacerbated by the rapid advancements in AI technology, which continually improve the coherence and relevance of generated texts. Moreover, the ability of AI to generate high-quality text necessitates a reevaluation of intellectual property policies. Current academic policies are often inadequate in addressing the ethical implications of AI-generated content and the need to clearly distinguish between human and AI contributions. There is an urgent need for updated academic policies that address these issues and ensure the ethical use of AI in academic writing [11].

Our study aimed to address two distinct gaps: the limitations of current academic policy frameworks, particularly regarding ethical AI use, and the shortcomings of traditional teaching methods in medical education, which often fail to effectively engage students and enhance academic performance. To bridge these gaps, we proposed the adoption of Project-Based Learning (PBL), a methodology designed to foster active learning and critical engagement with complex problems. PBL specifically addresses the issue of traditional teaching methods by engaging students in real-world projects that require practical application of knowledge, thus promoting critical thinking, problem-solving, and research skills. Furthermore, by encouraging students to take ownership of their projects and requiring transparent documentation of their processes, PBL directly addresses challenges related to ethical AI use, such as originality and authenticity in academic work. This dual focus makes PBL an effective strategy for cultivating both practical competencies and ethical awareness, equipping students with skills that are essential for their future medical careers and for navigating the evolving landscape of AI in education [12–17].

2 | Materials and Methods

A total of 179 students participated in this study. Of these, 71 students completed the course through literature review-based assessments, while 108 students engaged in a project-based approach focused on reviewing and analyzing specific healthcare topics. Participants were not randomly assigned to these groups but were part of intact classes. These students were in their twelfth semester of a single-step medical curriculum, having completed courses such as “Academic Writing,” “Introduction to Medical Research,” “Critical Thinking,” and “Biostatistics and Scientific Report.” Additionally, they were enrolled in clinical subjects and had exposure to clinical environments and clinical reasoning sessions. Both groups of students worked individually under mentorship to investigate relevant medical research questions, being responsible for designing the study, collecting and analyzing data, and presenting their findings. Given the similar backgrounds and abilities of the students in both groups, a sample size of 179 students was considered statistically relevant to capture the variability within the cohort, providing a reasonable level of confidence in the study outcomes.

The PBL group engaged in a structured process designed to foster active learning and critical thinking. The following steps outline the activities undertaken:

- Orientation and topic selection: students participated in an initial orientation session to understand the goals and expectations of the PBL approach. Under mentor guidance, each student selected a specific healthcare topic relevant to current medical challenges.
- Research planning: students developed a research plan, which included defining objectives, identifying data sources, and choosing appropriate methodologies. Mentors provided feedback to ensure feasibility and relevance.
- Data collection and analysis: students independently gathered data using various methods, such as literature reviews, surveys, or case studies. The collected data were analyzed using techniques discussed in prior coursework.
- Draft preparation and feedback: students prepared initial drafts of their findings. These drafts were reviewed in mentoring sessions, where constructive feedback was provided to refine their work.
- Final report and presentation: students revised their drafts based on feedback, submitted a final written report, and presented their findings to their peers and mentors for further discussion and evaluation.

These activities were designed to stimulate real-world research scenarios, helping students develop essential skills, including critical thinking, problem-solving, and research competence.

The Turnitin service, provided by Turnitin LLC, was used to collect and evaluate the students’ submitted papers. In this study Turnitin was employed to evaluate the originality of student submissions by calculating the similarity percentage of their content. Additionally, its AI detection capabilities were utilized to identify sections of text that may have been generated using AI tools. While these two features serve different purposes, they were both critical in ensuring academic integrity in this context.

The research reports were evaluated using a standardized writing rubric that assessed key aspects of academic performance, including research relevance, methodological rigor, data interpretation, originality, and clarity of writing. To ensure impartiality, evaluations were conducted by independent reviewers who were blinded to the participants’ group allocation. The rubric’s reliability and validity were established in prior academic studies [18], inter reliability in this study was confirmed with Cohen’s kappa (value = 0.89). These measures ensured a consistent and unbiased evaluation process.

The statistical methods employed in the study included a combination of descriptive analyses to examine differences in academic performance and similarity percentages between the two groups. Descriptive statistics summarized the characteristics of the student cohorts, including their academic performance scores and similarity percentages. Academic performance was measured using course grades, with measures such as averages,

means, and standard deviations providing a comprehensive overview of the data.

Ethical clearance was secured from the Bioethics International Committee of the Petre Shotadze Tbilisi Medical Academy (identification code: 20240503/01, Tbilisi, Georgia). All procedures adhered to the Helsinki Declaration of 1975, revised in 2013, with participants receiving comprehensive study information and providing written informed consent before inclusion.

3 | Results

Students engaged in PBL demonstrated significantly higher academic performance compared to those in traditional literature review-based assessments. The average academic performance score for PBL students was 82.5, compared to 66.5 for those in literature review-based assessments. PBL methodology resulted in lower similarity percentages, indicating greater originality in student submissions. The median similarity percentage for PBL was 4.5, compared to 13 for literature review-based assessments. Moreover, the data on AI involvement is available for literature review-based assessments only. For PBL activity it was indicated as an asterisk (*) and no percentage was attributed (Table 1). The lower similarity percentages in PBL indicate a reduced likelihood of academic misconduct and a greater emphasis on individual through processes in student work. The high percentages of AI involvement in literature review-based may suggest that students rely on AI tools for literature review and are compromising the originality of their assignments. The AI involvement data for PBL indicates that the methodological approach fosters authentic learning pathway.

4 | Discussion

Our findings revealed that Turnitin not only provided reliable similarity percentages but also assisted in detecting potential AI-generated content within student submissions. To enhance clarity, we have adopted the term “percentage of AI involvement” to describe the extent to which AI tools may have influenced the content. However, it is important to recognize the

limitations inherent in these detection methods. As noted in previous research [19, 20], AI detection tools can produce false positives, potentially leading to unwarranted accusations. Moreover, advanced techniques such as “paraphrasing attacks,” “prompting strategies,” and “adversarial techniques” [20] pose challenges to the reliability of these systems. Turnitin’s AI detection tool was effective in identifying most AI-generated content but struggled with more sophisticated texts. Our findings underscore the importance of integrating innovative educational methodologies to enhance academic performance and maintain academic integrity. PBL not only improve learning outcomes but also reduce the likelihood of AI-generated plagiarism by promoting original work and critical thinking [21–23]. The significantly lower similarity indices observed in PBL group indicate that these methods foster more authentic and original submissions compared to traditional literature review-based assessments [24]. To address the challenges posed by AI, we propose several assessment strategies:

1. **Oral examinations:** Incorporating oral presentations can mitigate the risk of AI-generated content by requiring students to demonstrate their understanding verbally (Chen et al., 2022). Oral exams provide a direct assessment of student knowledge and understanding, reducing the risk of AI-assisted cheating and promoting critical thinking and effective communication skills.
2. **Process-based assessment:** Emphasizing the writing process, including drafts and revisions, can help ensure the authenticity of student work. Process-based assessment involves evaluating the various stages of a student’s work [25], from initial drafts to final submissions. This approach discourages the use of AI tools for generating complete assignments, as students must demonstrate their progress and revisions over time. Reflective journals, where students document their thought process, challenges, and revisions throughout their project, can provide insights into the student’s learning journey and help identify areas where additional support may be needed.
3. **Ethical AI education:** Educating students on the ethical use of AI tools and their limitations can foster responsible use and enhance academic integrity. Ethical AI education should cover topics such as the ethical implications of AI, the potential biases in AI-generated content, and the importance of transparency when using AI tools (Nguyen

TABLE 1 | Academic performance and percentage of AI involvement for PBL and literature review-based assessments.

Statistical parameter	Literature review-based assessment	Project-based learning (PBL)
Average academic performance	66.5	82.5
Median academic performance	63.75	85
Standard deviation	4.812	5.800
Average Similarity Index (%)	12.62	4.17
Median Similarity Index (%)	13	4.5
Standard deviation (Similarity)	5.485	2.483
Average percentage of AI (%)	37.57	—
Median percentage of AI (%)	37.80	—
Standard deviation (AI)	3.29	—

et al., 2022). Students should be encouraged to view AI as a supplementary tool rather than a replacement for their own critical thinking and creativity.

Furthermore, PBL effectively promotes active learning and enhances academic performance [26]. By engaging students in real-world projects, PBL fosters critical thinking, collaboration, and practical research skills. The significant improvement in academic performance and lower similarity indices among PBL students highlight its effectiveness in promoting originality and reducing plagiarism. PBL encourages students to engage in real-world projects, providing hands-on experience that is crucial for developing research competencies and critical thinking skills [17]. This approach not only improves academic performance but also prepares students for future academic and professional challenges.

It has been revealed that combining written assignments with oral presentations and practical demonstrations ensures comprehensive evaluation of student knowledge [27]. The implementation of this practice has yielded positive results [28, 29]. The incidence of high-similarity submissions has decreased, and students have shown a greater understanding of the ethical use of AI tools. Additionally, the quality of student work has improved, reflecting a more genuine engagement with the learning material. Our study involving 179 students demonstrated that those engaged in PBL had higher average academic performance scores (82.5) compared to those in literature review-based assessments (66.5). This significant enhancement in academic performance underscores the potential of PBL in fostering critical thinking and practical skills essential for medical practice.

5 | Conclusion

The application by students of AI in academic writing presents significant challenges to academic integrity. However, by adopting innovative assessment strategies and promoting ethical AI education, institutions can uphold integrity while leveraging the benefits of AI technologies. PBL methodologies effectively enhance academic performance and promote originality in student work, making them valuable tools in modern medical education. Future research should focus on developing and refining these strategies to adapt to the evolving educational landscape.

6 | Limitations

This study, while providing valuable insights has several limitations that should be considered. Firstly, the sample size and diversity of the study are limited. The research involved 179 students from a single medical academy, which may not be representative of other institutions or student populations. Furthermore, a limitation of this study is the lack of random assignment of participants to the two groups. Instead, the groups consisted of intact classes, which may have introduced pre-existing differences in student characteristics, such as academic performance, motivation, or prior knowledge. The

study's duration was confined to a single semester. While the results indicate immediate benefits of PBL, the long-term effects on academic performance and integrity remain uncertain. This study acknowledges limitations regarding the accuracy and reliability of AI detection tools. While Turnitin was used to assess the “percentage of AI involvement,” previous studies have highlighted issues such as false positives and the potential for unjustified allegations [19, 20]. Additionally, the study did not account for bypassing techniques, future research should explore these aspects in greater depth to strengthen academic integrity measures. The study also did not include a direct control group for comparison purposes. While it compared the PBL approach with literature review-based assessments, having a control group that did not participate in either method would provide a clearer baseline to measure the effectiveness of PBL. The focus on medical students means that the findings may not be applicable to students in other academic disciplines. The unique nature of medical education, with its emphasis on practical and clinical skills, may influence the outcomes of PBL differently than in other fields of study. Further research is necessary to explore the impact of PBL across various academic disciplines to determine its broader applicability.

Author Contributions

Kldiashvili Ekaterina: conceptualization, methodology, project administration, writing – original draft, writing – review and editing. **Mamiseishvili Ana:** investigation, validation, formal analysis, writing – original draft. **Zarnadze Maia:** investigation, validation, formal analysis, writing – original draft.

Acknowledgments

Authors express their sincere appreciation to the students who participated in this study for their dedication and engagement.

Ethics Statement

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

No/Not applicable (this manuscript does not report data generation or analysis).

Transparency Statement

The lead author Kldiashvili Ekaterina affirms that this manuscript is an honest, accurate, and transparent account of the study being reported; that no important aspects of the study have been omitted; and that any discrepancies from the study as planned (and, if relevant, registered) have been explained.

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